

An Algorithm for Drawing Contours

There are many procedures for drawing the contours of a function but most of them proceed in the same basic way [32, 46, 63, 72, 83]. The function is evaluated at a collection of points in the plane, usually a rectangular grid, and then approximate contours are computed by interpolation using this evaluation.

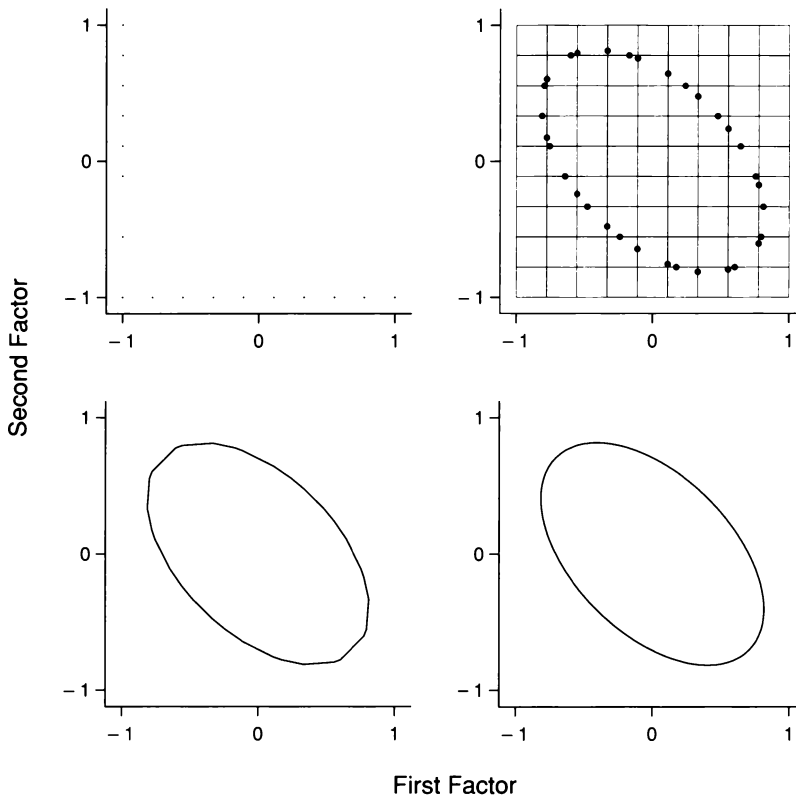
One algorithm for computing contours is illustrated in Figure 4.51, which shows the computation of the 0.5 contour of the function

$$g(u, v) = u^2 + v^2 + uv .$$

The actual contour is an ellipse centered at (0,0).

The first step in the algorithm is to evaluate g for a rectangular grid of points covering the region of study. A grid is shown in the top left panel of Figure 4.51; there are 10 rows of points and 10 columns, so altogether there are 100 points. The grid gives rise to a collection of horizontal and vertical line segments that connect the grid points. These segments are drawn in the top right panel of Figure 4.51. Consider all grid segments for which the function value at one endpoint is greater than 0.5, and the function value at the other endpoint is less than 0.5. Since our function is continuous, the contour for $g(u, v) = 0.5$ must cross each of these segments somewhere in the interior of the segment. If the function is 0.5 at an endpoint, then, of course, the contour also crosses the segment, but at the endpoint. In the top right panel of Figure 4.51, the *crossed segments* for $g(u, v) = 0.5$ are marked with dots.

The next step is to determine locations on crossed segments where the function is 0.5, or approximately so. If the function is equal to 0.5 at an endpoint, then the endpoint is such a location. If the function is greater than 0.5 at one endpoint and less at the other, we use linear interpolation to determine a location. A line is put through the two function values at the endpoints; the location is the position where the line is equal to 0.5. On the top right panel of Figure 4.51, the dots show the 0.5 locations that arise from this method.



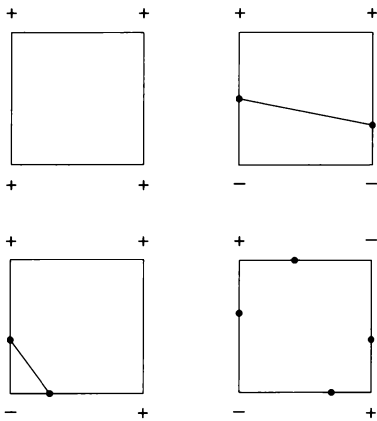
4.51 The figure illustrates one method for computing contours.

The final step in the contour algorithm is to connect the 0.5 locations of the crossed segments. A connection is illustrated in the bottom left panel of Figure 4.51. Doing the connection by computer might seem trivial, but it can be tricky because certain ambiguities can occur. One connection algorithm is given at the end of this section.

For most contour procedures, the only choice that needs to be made is the density of the grid. Greater density is typically better than less. If the grid is too coarse, more ambiguities can occur and the connection method can become confused. Also, contours that should be smooth can look jagged. For example, the contour in the bottom left panel of Figure 4.51 suggests the elliptical shape of the contour, but because the grid is coarse, there are unpleasant corners. A more sophisticated algorithm could round corners, but in this example, and in many others like it in practice, the simple solution is to increase the density of the grid. The same contour for a 50×50 grid is shown in the bottom right panel of Figure 4.51; the result is a curve that now appears smooth.

For the Record: A Connection Algorithm

One approach to connection is local in the sense that the connection of dots on a rectangle of the grid is not based on the locations of any other dots. Suppose the contour value is $c = 0$. Consider a rectangle with no dot at an endpoint of a segment. There are four possibilities, which are illustrated in Figure 4.52 — no dots, two dots on opposite sides, two dots on adjacent sides, and four dots. The figure shows the connection method for no dots and two dots. The four-dot case is ambiguous because there are several possibilities: two line segments with positive slopes connecting points on adjacent sides, two line segments with negative slopes connecting points on opposite sides, and so forth. And ambiguities can occur when dots are at endpoints.



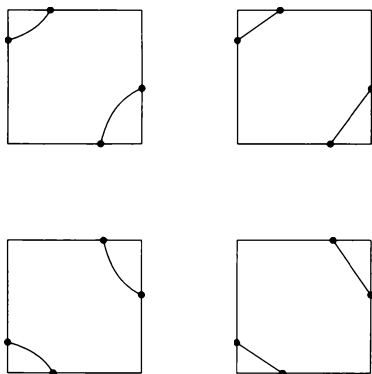
4.52 The figure illustrates the four possibilities for crossings when the contour level is zero and when the function is not zero at an endpoint. The connection method is shown except in the lower right panel, which is an ambiguous case.

Grosse developed an elegant method for resolving the ambiguous cases [46]. Consider first, the four-dot case in the lower right panel of Figure 4.52. Let

$$g(u, v) = a + bu + cv + duv$$

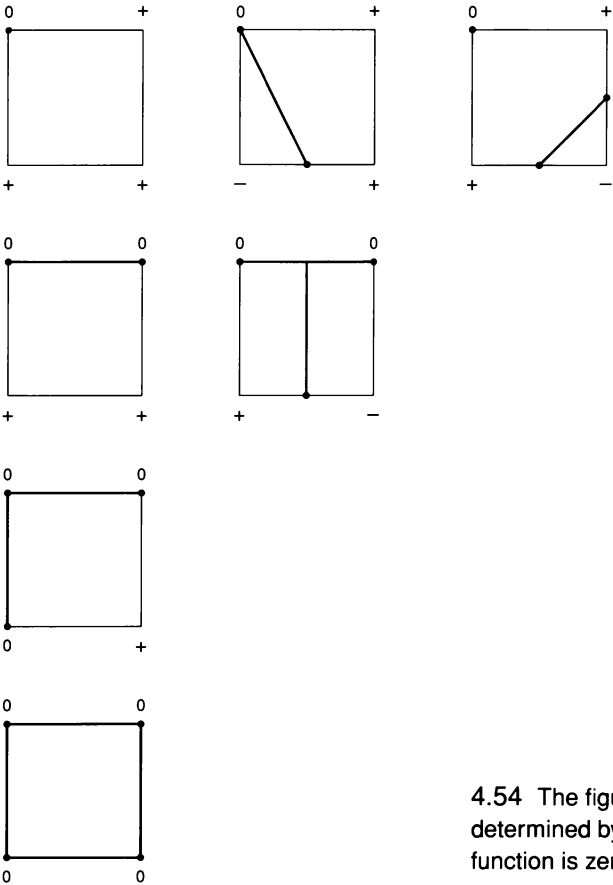
be the bilinear function put through the values of the function at the endpoints. The fit is exact since there are four parameters and four values being fitted. Suppose we used this bilinear function as the interpolant to determine the positions of the dots. The results would be the same as the method described above. In other words, as an interpolant, the bilinear function does linear interpolation along the segments of each rectangle.

But the bilinear is defined inside the rectangle. This means it has contours. And these contours, which are hyperbolas, have a special property: they cross the rectangle either as in the upper left panel of Figure 4.53 or as in the lower left. In Grosse's method, the bilinear contours are used to determine the connection. If they cross as in the upper left, the dots are connected as shown in the upper right. If they cross as in the lower left, the dots are connected as shown in the lower right. But here is the best part. The bilinear function does not need to be fitted to determine the connection. There is a very simple rule: the dots with the two smallest horizontal coordinates are connected, and the dots with the two largest horizontal coordinates are connected.



4.53 The figure shows how the connection is determined by Grosse's method for the ambiguous case of Figure 4.52.

Grosse’s method of bilinear approximation can also be applied when one or more values at the endpoints are zero. The possibilities, and the connections to which the method leads, are shown in Figure 4.54.



4.54 The figure shows how the connection is determined by Grosse's method when the function is zero at one or more endpoints.

4.9 *Level Plots of Surfaces*

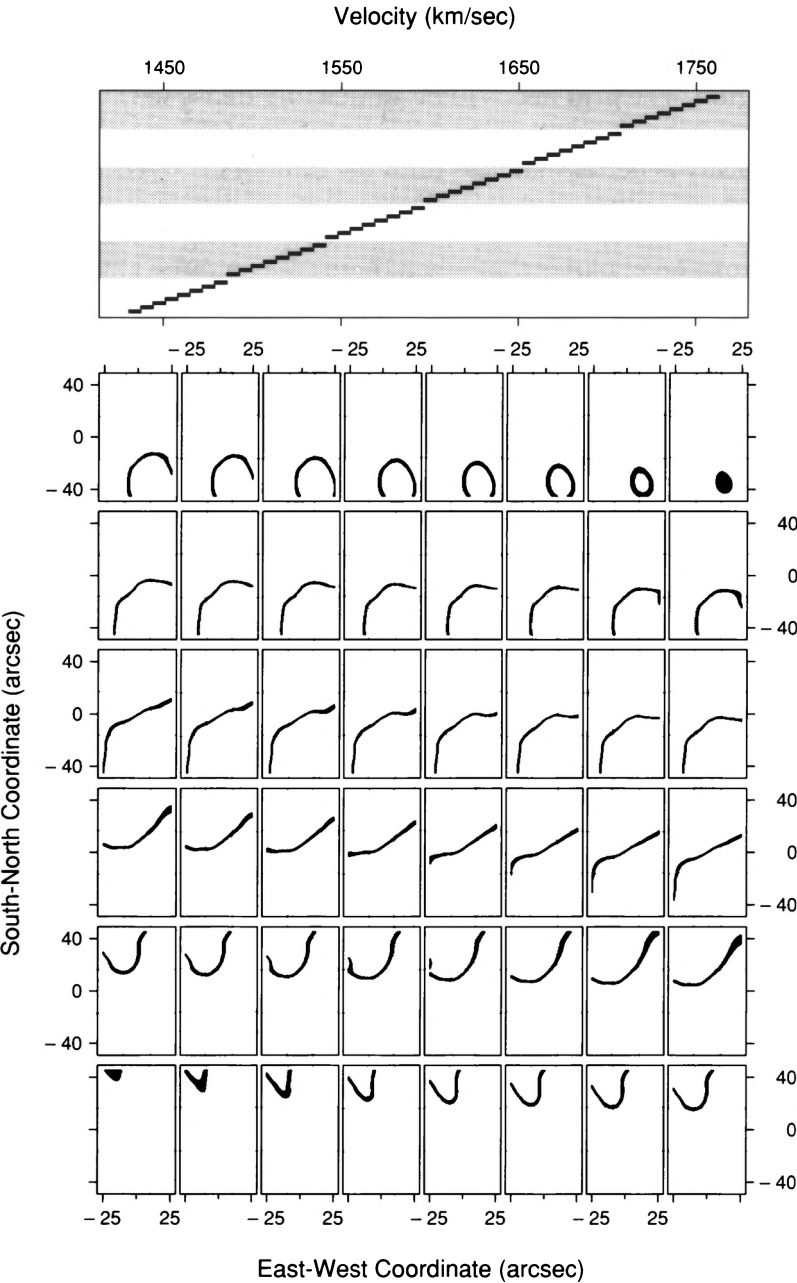
Contour plots can succeed. The contour plot of the velocity surface in Figure 4.50 is one example. But there are two problems. The first is fundamental to the method. If a surface is very nearly flat in some region of the factor space compared with the steepness in other regions, the definition of a contour as a curve becomes unstable. If the surface is flat in a region — that is, equal to a constant over a region with positive planar area — the contour for the constant value is no longer a curve. Consider the bottom panel of Figure 4.54. The function is zero at all four endpoints. The connection method gives an answer, a rectangle, but if the function is zero on and inside the rectangle, the ideal visualization would show a contour containing both the rectangle and its interior. Sophisticated contouring methods solve the problem by ignoring it; contours do not enter flat or nearly flat regions.

A second problem is one of visual perception. On a contour plot we cannot effortlessly perceive the order of the contours from smallest to largest. We must resort to the slow process of reading the contour labels. For a simple surface such as the velocity surface this suffices. But for more complicated surfaces with many peaks and valleys, it does not.

Another method solves both problems — a level plot. This method was introduced in Section 4.6 for displaying data, but it can also be used for displaying fitted surfaces. For functions, there are two versions of the plot. One, described in this section, renders the level regions by juxtaposing them in black and white; this is the same display method used for level plots of data. The second version, described later in Section 4.12, uses color and superposes the level regions.

A level plot in Figure 4.55 displays the loess fit to the galaxy data. First, the surface was evaluated on the same 101×181 grid used for the contour plot of the surface in Figure 4.50. The grid points and the values of the loess surface on the grid form a trivariate data set. The level plot of the surface is simply a level plot of these data, constructed as described in Section 4.6, but with one difference: instead of using levels from the equal-count method, the 48 levels are of equal length. The levels range from 1431 km/sec, the minimum value of the surface over the grid, to 1763 km/sec, the maximum surface value. The levels overlap only at their endpoints, but in other applications it can make sense to have greater overlap or to have no overlap and gaps between the levels.

Figure 4.55 shows the same behavior of the individual contours as the contour plot in Figure 4.50 — a change from upward convexity to downward convexity as velocity increases. But in Figure 4.55 we can also detect erratic behavior over small areas. As velocity increases through panels (3,2) to (6,2), the level regions in the vicinity of $(-20'', 20'')$ shift rapidly and become disconnected; this is an indication of a rapid local fluctuation. In panels (7,5) and (8,5) there is another rapid fluctuation near $(20'', -20'')$, but not as pronounced as the first. Figure 4.43, the graph of the measurement locations, shows that these two locations are at the ends of the 133° slit. The display of the data along the slits in Figure 4.45, and the display of the residuals along the slits in Figure 4.47, do not indicate any peculiar behavior on the 133° slit or any lack of fit. But the ends of the 133° slit do lie in regions with little other data, so the fluctuations are most probably simply the artifacts that result from applying loess to regions that are sparse in data.



4.55 A level plot displays the loess fit to the galaxy velocities. The aspect ratio of the level-region panels has been chosen to make the number of data units per cm the same on both scales.

On a level plot of a function, each level region is rendered by placing a dot at a grid point if the function value at the point lies in the level corresponding to that region. The grid needs to be sufficiently dense to produce level regions that characterize the actual level regions of the function; thus greater density is better than less until the limit of resolution of the display device is reached.

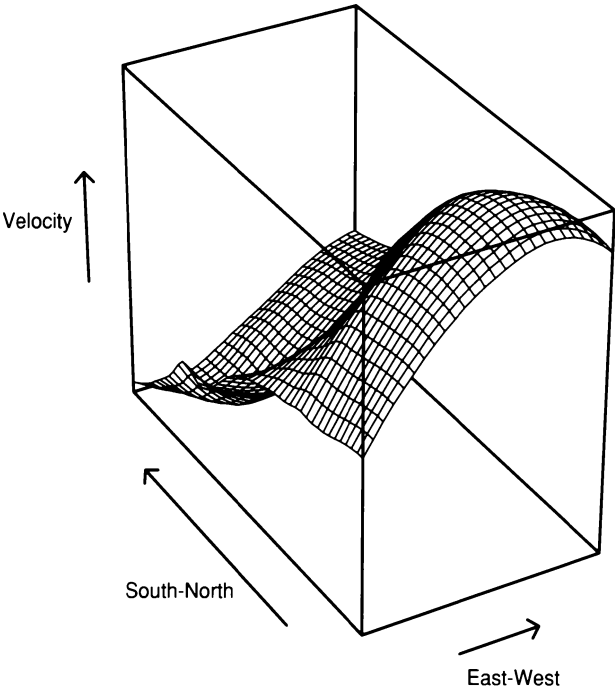
One important aspect of a level plot is that when there are contours that are well defined curves, we see them as the boundaries of the level regions. But when curves do not adequately describe the behavior of the function, we can still perceive the behavior. For example, if a function is constant over a region, the region appears as part of the level region for the level that contains the constant. A second important aspect of a level plot is that the arrangement of the panels in graphical order provides a mechanism for perceiving order that does not require table look-up. However, the drawback of rendering level regions on juxtaposed panels is a reduction in our ability to perceive the relative locations of level regions. In Section 4.12, superposition will be used.

4.10 3-D Wireframe Plots of Surfaces

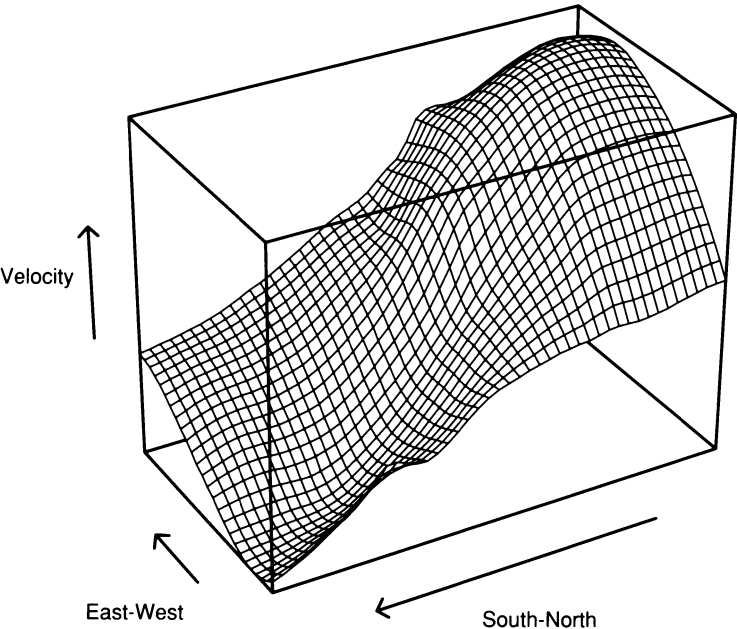
A 3-D plot of a fitted surface shows the surface as if it were an object in real 3-D space. Aspects of the display method allow us to see the surface-object in depth. Figures 4.56 to 4.59 are four 3-D plots of the velocity surface fitted to the galaxy data. The rendering in this case is a simple although effective method — a wireframe with hidden line removal [42, 79]. In Section 4.13 we will discuss more sophisticated schemes.

To carry out the wireframe rendering, the surface is first evaluated on a grid. The grid values together with the values of the surface form a set of points in three-dimensional space. The 3-D plot renders line segments that connect these points. Two aspects of the rendering, perspective and occlusion of features in the background by features in the foreground, provide the 3-D effect [68]. And the box drawn around the surface not only shows axes but contributes to our perception of depth; this is the reason for drawing all visible line segments of the box rather than just three to indicate axes.

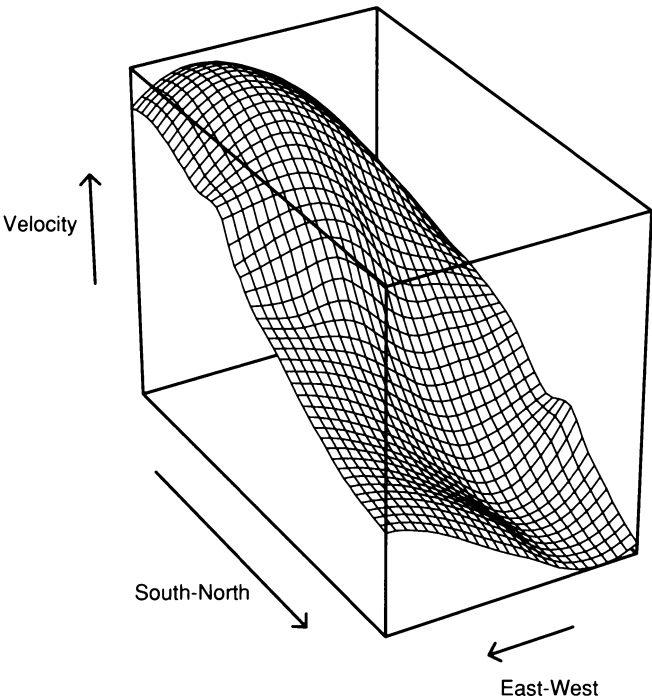
In Figures 4.56 to 4.59, the scale lines have arrows to indicate increasing values of the variables. There are no tick marks because scale reading, or table look-up, of numerical values from 3-D plots is often poor. To carry out table look-up for features detected on a 3-D plot, it is typically far more satisfactory to also display the surface by a level plot or by a contour plot, locate the detected features on the adjunct display, and use the scales of the adjunct display to carry out the table look-up.



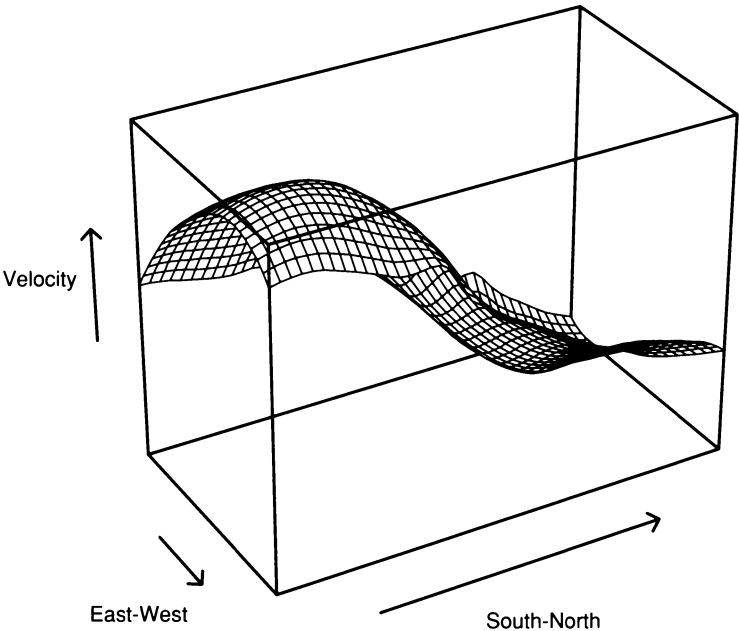
4.56 A 3-D wireframe plot displays the robust loess surface fitted to the galaxy velocities.



4.57 The surface in Figure 4.56 is shown again after a rotation of 90° about the velocity axis.



4.58 The surface in Figure 4.56 is shown again after a rotation of 180° about the velocity axis.



4.59 The surface in Figure 4.56 is shown again after a rotation of 270° about the velocity axis.

Rendering Choices

There are a surprisingly large number of choices to make in wireframe rendering. We must choose the grid density. We must choose the different orientations of the surface. We must decide whether to render the surface in perspective, or by an orthogonal view, a rendering of the surface as it would look if it were far away and studied by a powerful telescope. If we choose perspective, we must specify the viewing distance; as we move closer to the box, the effect of perspective increases. We must choose the ratios of the lengths of the three box dimensions, and the overall size of the display. None of the choices can be fully automated; they typically must be made iteratively based on successive trial renderings. The following discussion defines the issues and in some cases gives rough guidelines that can provide reasonable first attempts.

If the grid density is too small, surface features are lost; if it is too large, the white space of patches closes up, and depth perception is lost. Thus the goal is to choose the density just large enough to reflect important features and no larger. It can happen that there is no good choice. If the surface has fine structure that requires a grid so dense that, given the resolution of the display device, the patches are obscured, then we need to move to the more ambitious 3-D rendering described in the Section 4.13. For the 3-D plots of the velocity surface in Figures 4.56 to 4.59, the grid is 26×46 . This is less dense than the 101×181 grid used for the level plot of the velocity surface in Figure 4.55 and the contour plot of the surface in Figure 4.50; the denser grid is too dense for the 3-D plot.

Choosing several orientations is necessary in the common situation where no one view shows all features well; a single orientation that allows good perception of certain features can occlude other important features. But typically it is best to coordinate the different views by pointing the function axis in the same direction. This has been done in Figures 4.56 to 4.59. The result is views that form a revolution of the surface about the function axis. In addition, our perception is often enhanced by making the orientations of this revolution cover 360° in equal steps. In Figures 4.56 to 4.59, there are four orientations in steps of 90° . The coordination and the equal steps help us maintain our understanding of which axis is which on the 3-D plots.

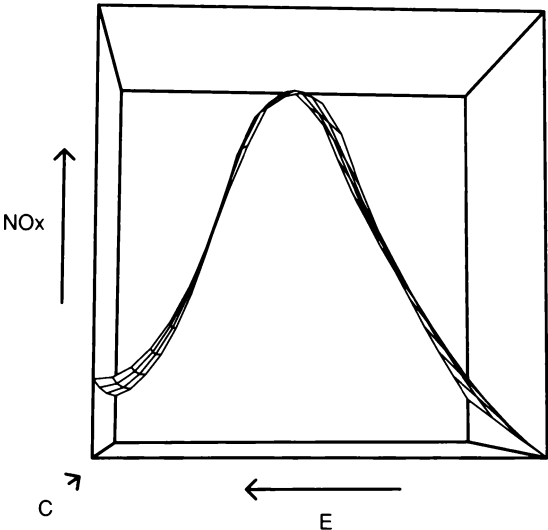
The relative lengths of the three box dimensions are important to perceiving the properties of the displayed surface. If the two factor variables have difference units of measurement, then the ratio of the two box lengths for the factors can simply be one. If, however, the units are the same — in particular, if the factors are spatial coordinates — then it is typically best to choose the factor lengths so that the number of data units per cm along each factor scale is the same. This has been done in Figures 4.56 to 4.59 because the factors are position on the celestial sphere. The relative length of the function, or response, box length needs to be chosen by trial and error; the choice determines a banking of surface slopes just as the aspect ratio of a two-variable graph banks the slopes of a curve. It is possible that there is a 45° principle, but the necessary research in visual perception has not yet been carried out to determine this.

A requisite amount of perspective is helpful for maintaining stable depth perception. Perspective creates foreshortening, a convergence of parallel lines that recede into the background. This foreshortening needs to be treated with great caution because, as a coming example will show, it is a distortion of the quantitative information. And the greater the amount of perspective, the greater the distortion.

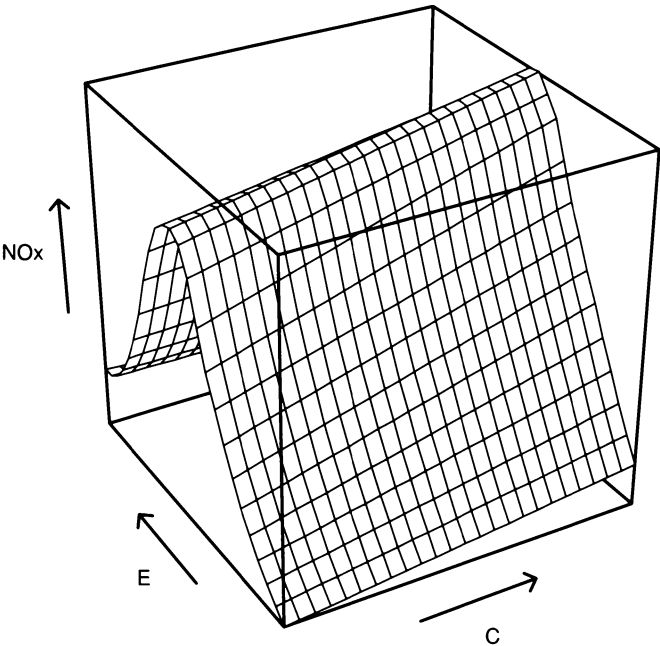
Orthogonal views eliminate the distortion of perspective; parallel lines moving from the foreground to the background stay a constant distance apart, and constant-size objects have a constant image size. But orthogonal views often result in other distortions. For wireframe renderings, it is often impossible without very focused attention on small regions of the display to determine front and back, so disconcerting reversals of foreground and background are common. For the more sophisticated renderings discussed in Section 4.13, where other visual cues such as shading convey depth, an orthogonal view provides conflicting information, and the algorithms of our visual system often respond by contorting the rendered object; for example, a rectangular box around the surface can appear nonrectangular. But an orthogonal view is needed for one visualization task that will be described in the next example.

Figure 4.60 is a perspective view of the loess surface fitted to the ethanol data. The surface appears to be a tunnel with arches of constant size. Figure 4.61 is another perspective view. The tunnel now appears to have arches whose size increases as C increases. Thus the two views convey conflicting conclusions about quantitative aspects of the surface. We know from the coplot of the fitted surface in Figure 4.13 that the arch sizes do in fact change, so we are fooled in Figure 4.60. For this figure, the arches expand in size from the foreground to the background. But the foreshortening due to perspective makes the actual image size of these arches decrease from front to back, and the decrease in size just matches the actual increase, so all arches are nearly visually coincident. To understand this phenomenon, it helps to consider that when we look into a tunnel with constant arch size, foreshortening makes the actual image size of the opening at the opposite end smaller than that of the opening at the near end.

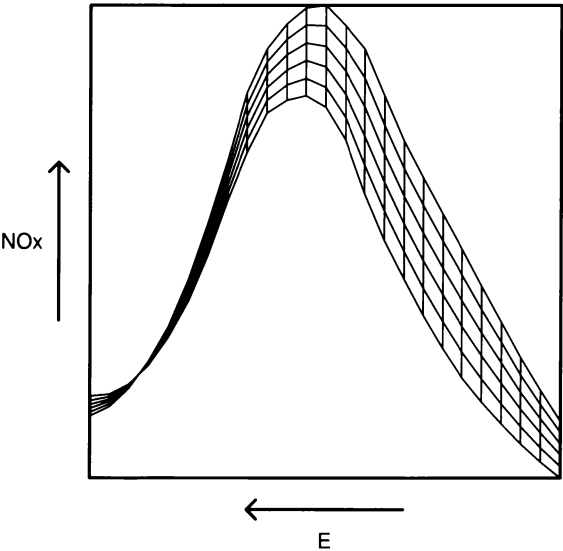
An orthogonal view is needed for looking along a direction parallel to one of the three axes to see the relative locations of the orthogonal projections of features onto the plane formed by the other two axes. A perspective view does not give orthogonal projections as Figure 4.60 has illustrated. Figure 4.62 shows an orthogonal view of the NO_x surface. The view is parallel to the C axis. Because there is no perspective distortion, we can trust what we see; for low values of E , the surface is nearly constant as a function of C , and for middle and large values of E , there is a change with C .



4.60 A 3-D plot displays the NO_x surface.



4.61 Another 3-D plot displays the NO_x surface.



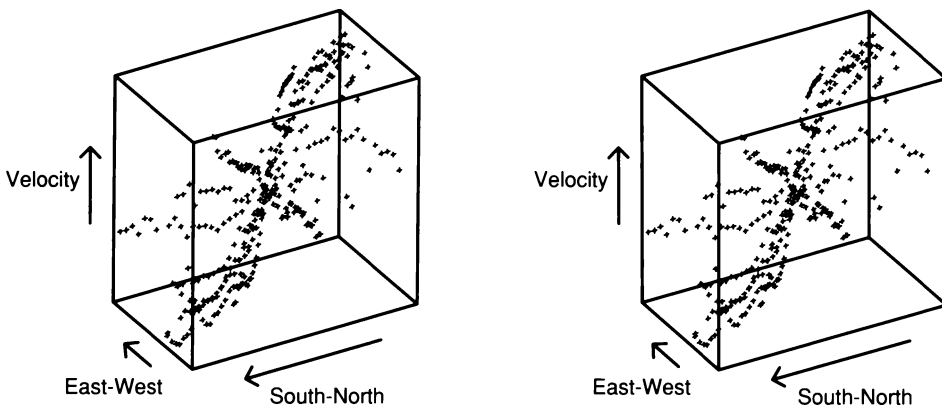
4.62 This 3-D plot of the NO_x surface uses an orthogonal view.

4.11 3-D Plots of Data: Stereo

The visual impression of depth that we get from occlusion and perspective foreshortening in 3-D plots of functions does not extend readily to data. If the number of observations is small, we can endow plotting symbols with depth cues such as changing symbol size and occlusion, but for even moderate amounts of data, the result is not reliable and too often results in an inscrutable mess despite our best efforts.

Stereo viewing is a far more satisfactory method for 3-D rendering of data. Stereopsis is a powerful mechanism for depth perception. Figure 4.63 shows a stereogram of the galaxy data. For those without a stereo viewer, Chapter 2 has instructions on how to trick the eyes into fusing the two images. The method is not easy to master, but practice helps. The overall behavior of the galaxy velocities — an increase in going from the northeast corner to the southwest — is lucidly conveyed.

Motion is another powerful cue for depth perception. For trivariate data, a rotating point cloud on a computer screen provides excellent depth perception [41]. Furthermore, methods of direct manipulation allow us to control the orientation of the point cloud in a natural, perceptual way. Sadly, we cannot even approximate rotation in the static environment of a book because the animation is critical to the perception.



4.63 A stereogram displays the galaxy velocities.

4.12 *Level Plots of Surfaces with Superposed Color*

The visual displays so far presented in the book have been images created by line segments and symbols rendered in black and gray. But computer graphics allows more sophisticated rendering [42, 79]. Among the capabilities is color, which can be used to good purpose to produce level plots.

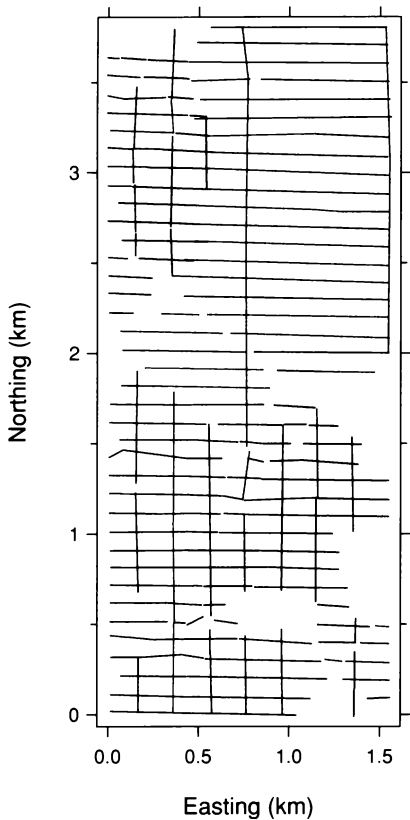
Soil Data

How things get measured is a fascinating topic. And a critical one, of course. In many domains of science, brilliance and great ingenuity abound in measurement techniques. This intellectual effort is crucial because measurement is the foundation of scientific enquiry.

Suppose we had to measure the electrical resistivity of soil. Would we simply insert two electrodes in the ground and measure the current flowing between them? That would work, but suppose now that we had to survey the resistivity at ten thousand locations over a region of one square kilometer. Just to make the problem a little more difficult, suppose we had to run a business doing this and make a profit. Resistivity measurements can be used to infer soil salinity, a vital matter for irrigated farm land. It is unlikely that we could economically stick electrodes in the ground ten thousand times.

A solution was found by the Salinity Surveys division of TESLA-10, a company in Western Australia where irrigation is necessary for farming. Their instrument, the EM31, has a transmitter coil that induces a current in the earth. The current generates a magnetic field whose intensity is measured by the EM31. The instrument is put on a trailer and towed around by a jeep, which stops at regular intervals and takes a measurement. Thus soil resistivity is measured economically by an instrument that never touches the ground.

Figure 4.64 shows the locations of 8641 measurements made by the EM31. Each measurement location is graphed by a dot, but the density of the dots is so great that they merge to form curves, the measurement tracks. The vertical scale is labeled “northing” and the horizontal is labeled “easting”. These are just names for a local coordinate system set up for convenience; they have nothing to do with compass directions. For the most part, the vehicle towing the EM31 tracked along the easting axis, but it also made a few runs along the northing axis.



4.64 The display shows the measurement tracks for the soil data. The aspect ratio has been chosen to make the number of data units per cm the same on both scales.

Spatial Data

The soil data are spatial data; the factors are geographical location. Spatial data, like time series, often have different components of variation. Time series have time components, and spatial data have spatial components. The spatial components can be trends, or long-distance variation; they can be slow oscillations, or middle-distance variation; or they can be rapid ups and downs, or short-distance variation. The galaxy measurements analyzed in previous sections are also spatial data: velocities at spatial locations on the celestial sphere. The visualizations showed that the variation in velocity consists of a broad trend plus nearly random variation about the trend. Shortly, the visualization of the soil data will show many spatial components, from long-distance to short-distance variation.

Improvisation

We can improvise, exploiting the measurement tracks to visualize the raw data. In Figure 4.65, resistivity is graphed against the northing coordinate for the eight tracks that run along the northing axis. On the display, convention has been ignored, and the factor, the northing coordinate, is graphed on the vertical axis as it is in Figure 4.64. This makes it easier to match the panels of Figure 4.65 with the tracks in Figure 4.64. As the numbers at the tops of the panels increase, the tracks move from left to right along the easting axis. In Figure 4.66, resistivity is graphed against the easting coordinate for the 40 tracks that run along the easting axis. The easting coordinate is on the horizontal scale, also to conform with Figure 4.64; as the panel numbers increase, the tracks move from bottom to top in Figure 4.64.

Figures 4.65 and 4.66 show that the soil data have many spatial components of variation. Rapid local variations and slow oscillations are quite evident. There are also broad trends; for example, panels 29 to 36 of Figure 4.66 reveal an incline running along the entire easting axis.